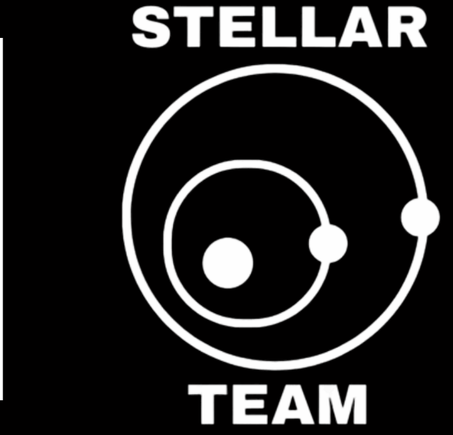
Análise da janela espectral na caracterização de Estrelas Anãs M com Deep Learning



Josué Araújo

Trabalho base: Bello-Garcia+, 2023

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Astronomy & Astrophysics

The CARMENES search for exoplanets around M dwarfs

A deep transfer learning method to determine $T_{\rm eff}$ and [M/H] of target stars★

A. Bello-García[1], V. M. Passegger[2,3,4,5], J. Ordieres-Meré[6], A. Schweitzer[4], J. A. Caballero[7], A. González-Marcos[8], I. Ribas[9,10], A. Reiners[11], A. Quirrenbach[12], P. J. Amado[13], V. J. S. Béjar[2,3], C. Cifuentes[7], Th. Henning[14], A. Kaminski[12], R. Luque[13,15], D. Montes[16], J. C. Morales[9,10], S. Pedraz[17], H. M. Tabernero[18], and M. Zechmeister[11]

ABSTRACT

The large amounts of astrophysical data being provided by existing and future instrumentation require efficient and fast analysis tools. Transfer learning is a new technique promising higher accuracy in the derived data products, with information from one domain being transferred to improve the accuracy of a neural network model in another domain. In this work, we demonstrate the feasibility of applying the deep transfer learning (DTL) approach to high-resolution spectra in the framework of photospheric stellar parameter determination. To this end, we used 14 stars of the CARMENES survey sample with interferometric angular diameters to calculate the effective temperature, as well as six M dwarfs that are common proper motion companions to FGK-type primaries with known metallicity. After training a deep learning (DL) neural network model on synthetic PHOENIX-ACES spectra, we used the internal feature representations together with those 14+6 stars with independent parameter measurements as a new input for the transfer process. We compare the derived stellar parameters of a small sample of M dwarfs kept out of the training phase with results from other methods in the literature. Assuming that temperatures from bolometric luminosities and interferometric radii and metallicities from FGK+M binaries are sufficiently accurate, DTL provides a higher accuracy than our previous state-of-the-art DL method (mean absolute differences improve by 20 K for temperature and 0.2 dex for metallicity from DL to DTL when compared with reference values from interferometry and FGK+M binaries). Furthermore, the machine learning (internal) precision of DTL also improves as uncertainties are five times smaller on average. These results indicate that DTL is a robust tool for obtaining M-dwarf stellar parameters comparable to those obtained from independent estimations for well-known stars.

Key words. methods: data analysis – techniques: spectroscopic – stars: fundamental parameters – stars: late-type – stars: low-mass

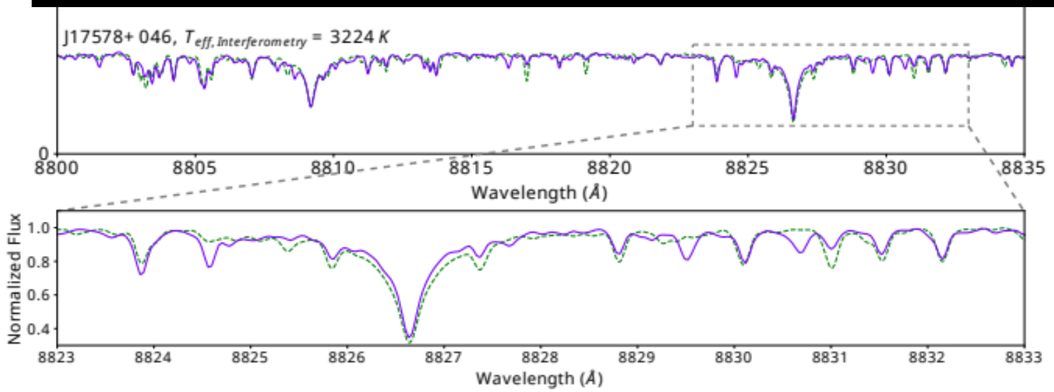
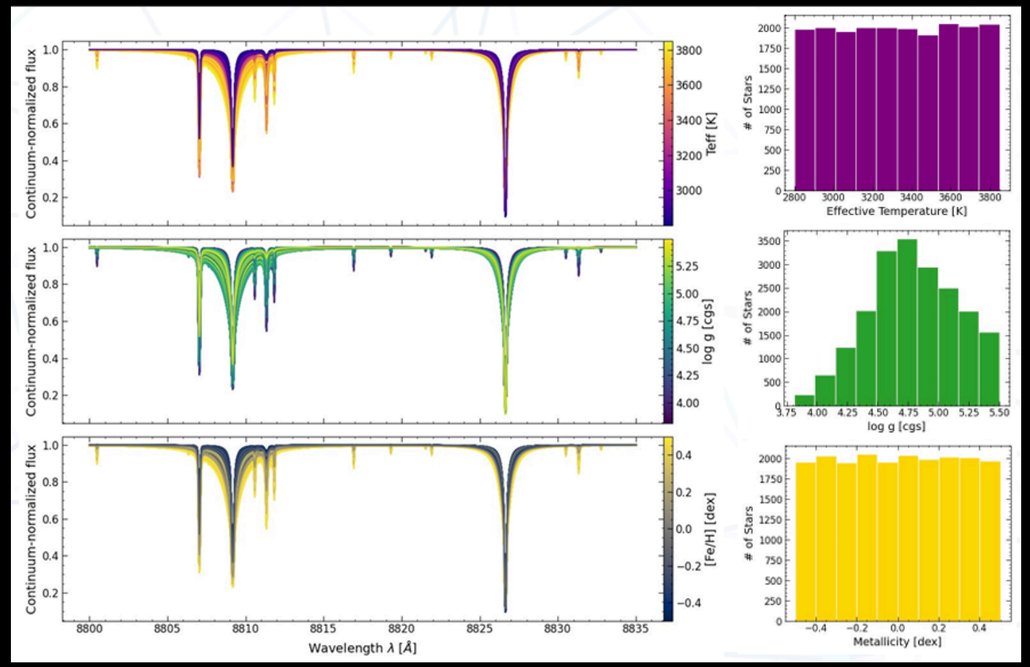
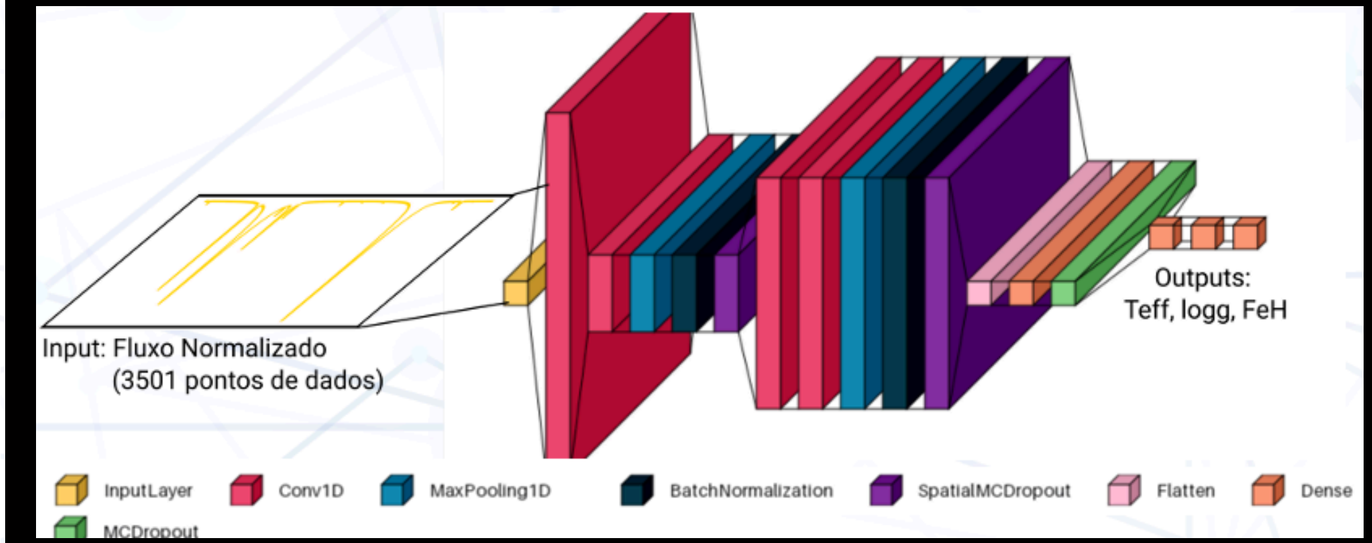


Fig. 1. Observed vs. synthetic fluxes (8800–8835 Å) for stars with interferometric $T_{\rm eff}$. Solid rainbow: CARMENES normalized template spectra. Dashed green: PHOENIX normalized synthetic spectra. Stars are sorted by decreasing $T_{\rm eff}$ from Schweitzer et al. (2019), with interferometrically derived $T_{\rm eff}$ indicated above them. Bottom panel: zoom-in detail of one representative spectrum.

Caracterização de Anãs M com Deep Learning





$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y}_i)^2} \qquad \text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \qquad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

	R^2	MAE	MSE	RMSE
Teff [K]	0.999	6.418	72.357	8.506
logg [cgs]	0.999	0.010	0.000	0.013
FeH [dex]	0.998	0.010	0.000	0.014

Tabela 1

Expansão do trabalho

J/A+A/627/A161/tableb1 — CARMENES stars multi wavelength measurements (Passegger+, 2019) — 2019A&A...627A.161P — ReadMe+ftp

Basic information of investigated M dwarfs *(282 rows)*

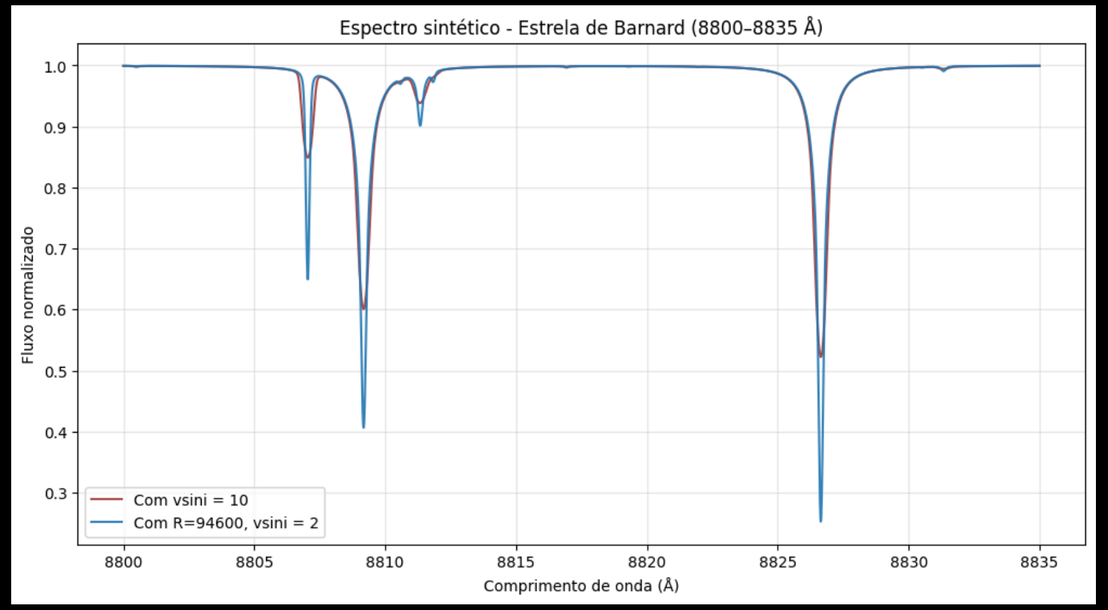
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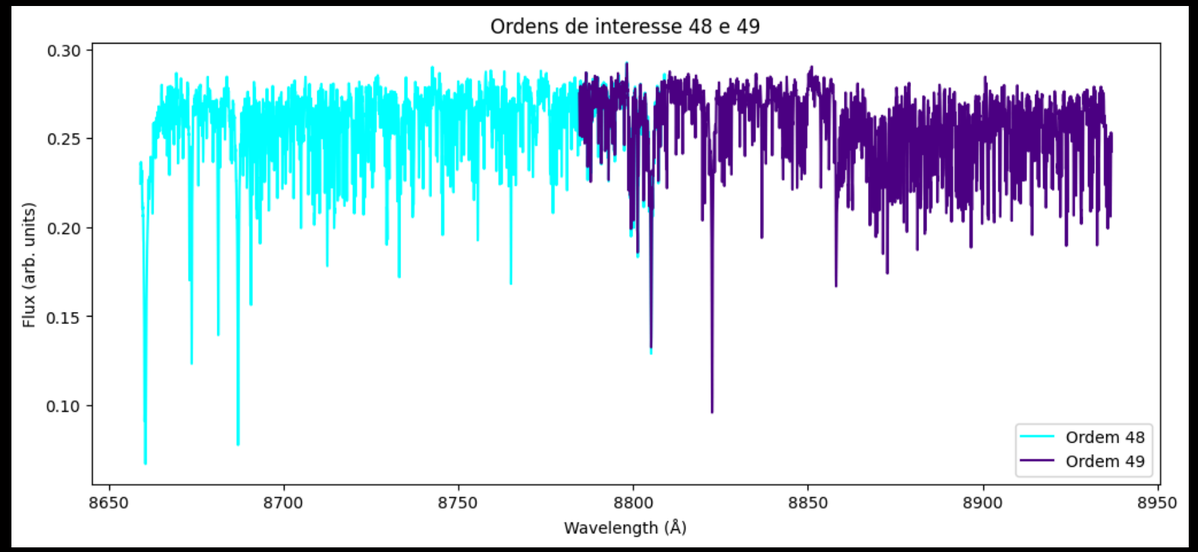
Full	Karmn	Name	GJ	RAICRS "h:m:s"	DEICRS "d:m:s"	SpType	l	vsini km/s	Age Gyr	factive	SimbadName	_RA.icrs "h:m:s"	_DE.icrs "d:m:s"
1	J17578+046	Barnard's Star	699	17 57 47.67	+04 44 16.7	M3.5V	<	2.0	8.00	0	Barnard's Star	17 57 47.67	+04 44 16.7

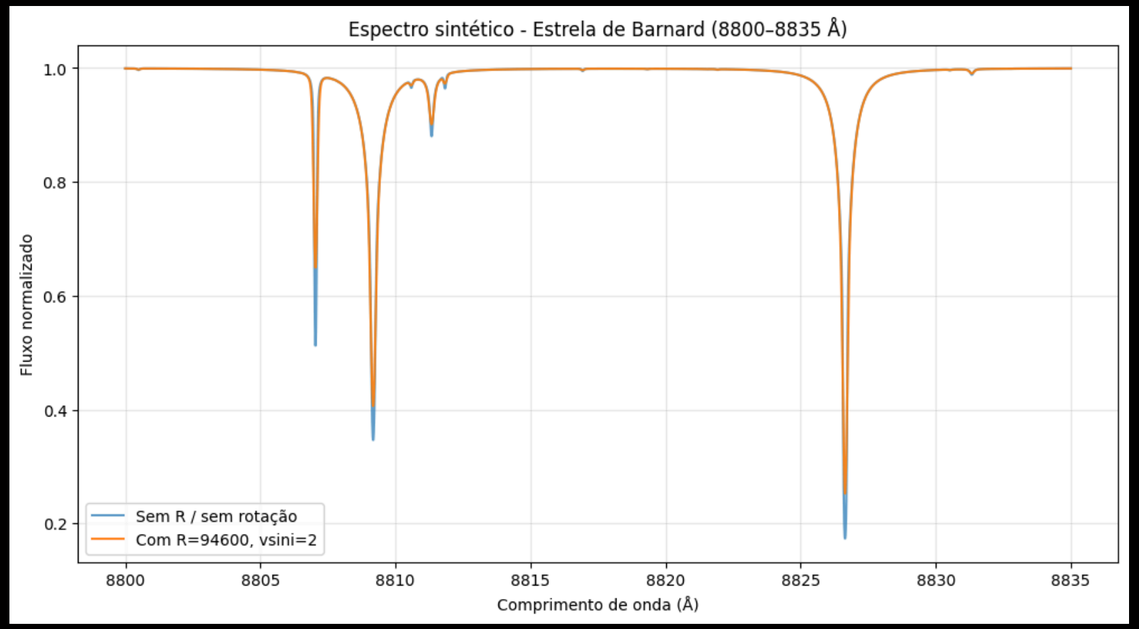
J/A+A/627/A161/tableb2 — CARMENES stars multi wavelength measurements (Passegger+, 2019) — 2019A&A...627A.161P — ReadMe+ftp

Basic astrophysical parameters of investigated M dwarfs in multiple wavelength ranges (corrected version) *(282 rows)*

Full	Karmn	TeffVN K	e_ K	loggVN [cm/s2]	e_ (...)	[Fe/H]VN [-]	e_ [-]	TeffN K	e_ K	loggN [cm/s2]	e_ (...)	[Fe/H]N [-]	e_ [-]	TeffV K	e_ K	loggV [cm/s2]	e_ (...)	[Fe/H]V [-]	e_ [-]
1	J17578+046	3243	54	4.96	0.06	-0.09	0.19	3278	56	4.93	0.04	-0.13	0.16	3238	51	4.96	0.04	-0.14	0.16







Problemática e nova perspectiva

“Moreover, Passegger et al. (2020) constructed several neural network models for different spectral regions, finding that the results for all regions are comparable, but that the region 8800–8835 Å gives the smallest validation error. Therefore, we also adopted this strategy and only use that region.”

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ABSTRACT

Existing and upcoming instrumentation is collecting large amounts of astrophysical data, which require efficient and fast analysis techniques. We present a deep neural network architecture to analyze high-resolution stellar spectra and predict stellar parameters such as effective temperature, surface gravity, metallicity, and rotational velocity. With this study, we firstly demonstrate the capability of deep neural networks to precisely recover stellar parameters from a synthetic training set. Secondly, we analyze the application of this method to observed spectra and the impact of the synthetic gap (i.e., the difference between observed and synthetic spectra) on the estimation of stellar parameters, their errors, and their precision. Our convolutional network is trained on synthetic PHOENIX-ACES spectra in different optical and near-infrared wavelength regions. For each of the four stellar parameters, $T_{\rm eff}$, $\log g$, [M/H], and $v \sin i$, we constructed a neural network model to estimate each parameter independently. We then applied this method to 50 M dwarfs with high-resolution spectra taken with CARMENES (Calar Alto high-Resolution search for M dwarfs with Exo-earths with Near-infrared and optical Échelle Spectrographs), which operates in the visible (520–960 nm) and near-infrared wavelength range (960–1710 nm) simultaneously. Our results are compared with literature values for these stars. They show mostly good agreement within the errors, but also exhibit large deviations in some cases, especially for [M/H], pointing out the importance of a better understanding of the synthetic gap.

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A questão científica central vai ser: Dentre um pequeno conjunto de janelas candidatas, qual delas melhor recupera os parâmetros estelares escolhidos em um benchmark sintético do Korg?

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- “Modelos sintéticos, embora muito utilizados, apresentam discrepâncias em relação a espectros observacionais (denominadas gap sintético), cujo perfil depende do código de síntese adotado. ”
- **“Propomos investigar, por meio de Deep Learning, como a eficiência de janelas espectrais varia entre diferentes domínios sintéticos, avaliando a generalização de regiões consideradas ótimas no código PHOENIX e identificando as mais eficientes no domínio do Korg (código de síntese baseado em regime 1D LTE). ”**
- *“Como perspectiva, a ampliação da complexidade dos espectros gerados e a análise de diferentes janelas espectrais permitirão avaliar sua eficiência no domínio do Korg em cenários mais realistas, permitindo assim investigar sua robustez e contribuir para a compreensão do gap sintético. ”*

Identificação de janelas espectrais robustas para a inferência de parâmetros estelares em anãs M e suas implicações para a caracterização de exoplanetas

Regiões do espectro não carregam a mesma quantidade de informação astrofísica, e algumas podem ser mais sensíveis a limitações do modelo sintético do que outras, synthetic gap. Como consequência, diferentes escolhas de janela espectral podem produzir diferentes níveis de precisão e robustez na inferência de parâmetros estelares. Sendo necessário investigar quais janelas espectrais oferecem melhor desempenho e maior confiabilidade para a recuperação desses parâmetros, reduzindo vieses sistemáticos que se propagam para a caracterização de exoplanetas.

Como objetivo: **Implementar uma CNN 1D para estimar parâmetros estelares de anãs M a partir de espectros sintéticos e investigar a eficiência de diferentes janelas espectrais na recuperação de Teff, log g e [Fe/H], com foco na caracterização robusta de estrelas hospedeiras de exoplanetas.**

Basicamente gerar um debate de como o erro na teff muda interpretações de luminosidade, zona de habitabilidade, log g no tamanho e densidade e, principalmente, na [Fe/H] para composições, frequência de planetas, etc.

2D - Stellar Abundances and Planetary Properties

Como expansão:

- Comparar Korg vs PHOENIX para saber se a melhor janela depende do modelo sintético?
- Se a janela muda, isso pode alterar conclusões sobre exoplanetas?